BSc (Hons) in **Information Technology Specialized in AI**

Faculty of Computing

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Practical 01

> **Objective & Lecture Mapping:** Recall that an intelligent agent acts within a continuous loop: Sensors generate percept sequences, and Actuators execute actions to maximize a Performance measure. Use this sheet to execute practical modifications and incrementally evaluate your design against Lecture 01 and 02 theory (from basic recall to analytical classification).

Part 1: Code Analysis & PEAS Evaluation

Task 0 : Setting up and getting the starter Code

1. Go to the Git Hub Repo. Create a Fork of this Git Repo to your Personal Github Profile. Open it using your favourite IDE and follow the below instruction to complete the practical. The base code is in the "master" brach of the repo.

Task 1.1: The Environment State (__init__)

1. (Remember) List the four components of the PEAS framework discussed in Lecture 01.

Performance measure
Environment
Actuators
Sensors

2. (Understand) Look at the `VisualGridHuntGame.__init__` method. Which specific variables in this code represent the physical "Environment (E)" state?

In visual_grid_game.py's __init__ method, the variables self.width, self.height, self.walls, self.food_positions, self.opponents, and (after our changes) self.toxic_traps all collectively represent the physical state of the grid environment.

3. (Analyze) Based on the variables initialized (specifically `self.opponents`), classify this baseline environment as "Single-Agent" or "Multi-Agent". Briefly justify your choice.

This is a Multi-Agent environment. The presence of self.opponents which actively move and change the state of the game (and can cause a collision that heavily impacts the agent's score) means our agent is not operating in isolation. It must

Task 1.2: The Perception Subsystem (get_percept)

4. (Remember) Define what a "percept sequence" is according to Lecture 01.

A percept sequence is the complete history of everything the agent has perceived from its sensors up to the current moment in time.

5. (Understand) Which component of the PEAS framework does the `get_percept()` method represent in our code?

This represents the Sensors component of the framework.

6. (Evaluate) Based on the exact dictionary returned by `get_percept()`, is this environment "Fully Observable" or "Partially Observable"? Explain why based on what the agent can and cannot see.

environment is Partially Observable. While the agent is given its own position and the exact positions of opponents, it is not given the coordinates of all the walls or food_positions. It only receives boolean flags ('smells_food', 'hit_wall') about its current position, meaning it cannot see the full layout of the maze and must explore to discover it.

Task 1.3: Action Execution (execute_action)

7. (Understand) When `execute_action()` deducts points for hitting a wall, which PEAS component is being actively updated?

The Performance measure (represented by self.score).

8. (Evaluate) Why is it structurally important that this metric evaluates changes in the external environment state (hitting a wall) rather than the agent's internal processing effort?

Performance measures must evaluate the actual, objective success of the agent in the environment (e.g., getting food, not hitting walls). If we measured internal processing effort (like how many loops the agent ran or how much memory it used), an agent could maximize its "score" simply by wasting computational resources without achieving the actual goal.

Part 2: Guided Modification & Deep Theory Mapping

> Follow the implementation instructions to inject a new hazard, then answer the questions to map your code changes back to the theoretical nature of the environment.

Step 2.1: Extending the Environment Initialization (__init__)

1. **Implementation Instructions:** Declare a new trap collection attribute (e.g., `self.toxic_traps = set()`) inside `__init__`. Populate it with coordinate tuples safely avoiding position `(0, 0)`, walls, and food.

9. (Understand) If you successfully add `self.toxic_traps` to the environment but intentionally hide this data from the agent's sensors, how does this specifically alter the environment's "Observability" classification?

It makes the environment Partially Observable. Because the agent can't see or sense the traps, it is basically walking blind. It only knows a trap exists after it steps on it and loses points, making the game much harder to predict.

Step 2.2: Updating the Perception Subsystem (get_percept)

1. **Implementation Instructions:** Locate `get_percept(self)` and add a new boolean sensor key (e.g., `'smells_toxin': tuple(self.agent_pos) in self.toxic_traps`) to the dictionary returned to the agent loop.

10. (Analyze) By adding `'smells_toxin'`, you expand the agent's percept sequence. Explain how this specific sensor helps the agent act with "Rationality" (maximizing expected utility).

A "rational" agent wants to get the highest score possible. If it can smell the toxin, it knows exactly why its score dropped. This allows the agent to learn from its mistake, remember where the trap is, and avoid stepping there again in the future to protect its score.

Step 2.3: Modifying Action Execution & Visual Rendering (execute_action & draw_grid)

1. **Implementation Instructions:** In `execute_action`, check if the updated `self.agent_pos` intersects with `self.toxic_traps` to decrement `self.score`

by 15 points. In `draw_grid`, iterate through traps to render custom purple shapes on the canvas.

11. (Remember) You programmed a severe score penalty for stepping on a trap to avoid metric exploitation. What was the classic "Vacuum World Exploit" example discussed in Lecture 01 that illustrated the danger of faulty metrics?

robot vacuum that gets +1 point every time it cleans up a piece of dirt. A sneaky robot might learn to dump its own dirt onto the floor and immediately vacuum it back up, doing this over and over to get endless points. This example shows what happens when you give an AI a bad scoring system it finds a way to cheat instead of actually doing the job.

Finalize and Lock Lab 01 Analytical Evaluation

FACULTY OF COMPUTING